

PROJECT IDEA

Utilizing sensing technology to predict potential impediments faced by an autonomous Vehicles

In this project, the aim is to use sensing technology to better understand the topology of road networks and vivid objects involved within a driving environment such as road signs, lane markers, other vehicles and terrain characteristics. The insights derived from this data will be instrumental in optimizing the training of existing and future autonomous vehicle software, enhancing their decision-making capabilities across diverse scenarios. This will contribute significantly to making autonomous driving more reliable and efficient.

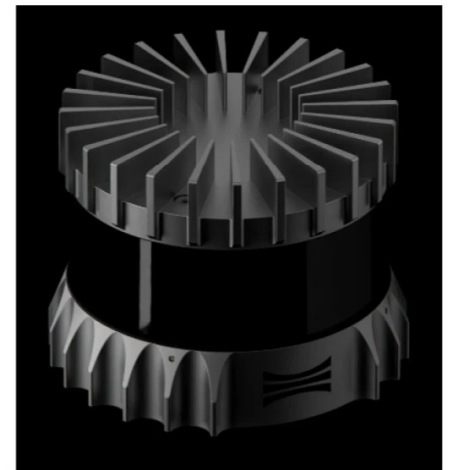
Currently, it is legal in Canada  to test autonomous vehicles on public roads. In the US, autonomous vehicles can be driven, provided an operator is present at the steering wheel. However, these undermines the core objective of autonomous driving. Therefore, upgrading current decision-making models in these vehicles is crucial, underscoring the necessity of this project.



SENSORS

Amongst all others, **LiDAR** sensing technology has been chosen for this project due to its accuracy and reliability compared to other sensing technologies.

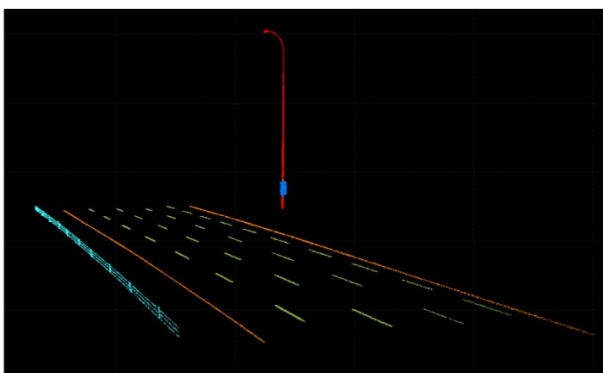
Specifically, the **OS1 model** of the **Ouster LiDAR sensor** series has been chosen for its exemplary features such as a max range of **200m** of detection, **128 channels** of resolution and **5.2 Million** samples per second capturing capacity.



FUNCTIONALITY

The pathway to this project includes building a **Deep Learning model** that is able to identify the diverse elements present in a driving environment based on **Semantic segmentation**, i.e. classifying the objects based on groundtruth data training and surrounding pixel state.

The equipment used to record the "**point-cloud**", i.e. is the LiDAR data is called a **Mobile Laser Unit (MLU)** and this unit consists of two more sensors called **Inertial Measurement Unit** and **Global Navigation Satellite Systems (GNSS)**. The output from all these sensors is integrated by the software called a transformer model and the results are used to analyze the topology of the driving environment.



"Self-driving cars are the natural extension of active safety and obviously something we should do." ~ Elon Musk

Research Progress

Transformer V2 model

So far, we have built a **transformer model** that is capable to discern various road object such as **Road Lanes, Traffic Signs, Concrete Barriers, Light Poles, Vegetation** and more. Latest results demonstrate a proprietary success rate compared to previous models based on sample data, with the added capability of annotating the LiDAR point cloud based on **semantic segmentation**.

This essentially pre-process the LiDAR data for autonomous vehicles enabling them to interpret the LiDAR sensor data and the annotation classifying the data into groups such as **road boundary, lane width, terrain condition, traffic signage** etc.

